MOHAMMED HOSSAM

Cloud DevOps & AI Engineer | Aspiring AI Product Manager
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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Computer Engineering graduate with a cloud and DevOps foundation (AWS, Terraform, Docker, Kubernetes, GitOps CI/CD) who also builds AI systems: an AI language-tutor prototype using an open-source LLM, a RAG pipeline and a vector database, served through FastAPI. Looking to learn AI product management on a real agentic product, turning customer and developer needs into well-scoped features while understanding the whole system from deployment to end user. Fast, self-directed learner with an original product concept for AI-driven language learning.

EDUCATION

Bachelor of Computer Engineering, Suez Canal University, Egypt *Sep 2020 – Jul 2024 | Grade: Very Good*

- Coursework: Secure Software Development, Database Systems, Operating Systems, Embedded Systems Security
- Completed projects in full-stack web application development and cloud engineering

SKILLS

- **AI & LLM:** Python, FastAPI, RAG pipelines, embeddings and vector search, LLM integration (OpenAI API, local models via Ollama: Llama 3, Mistral), prompt engineering
- **Product:** Product concept and design (AI language tutor), agile methodology
- **Cloud & DevOps:** AWS, Infrastructure as Code (Terraform); Docker, Kubernetes, Docker Swarm, Docker Compose; CI/CD with Jenkins, GitHub Actions, Argo CD; monitoring with Prometheus and Grafana
- **Systems & Networking:** Linux administration (nginx, tomcat), user management, backups, log analysis, firewall and security rules, troubleshooting, network infrastructure and security
- **Database systems**: MySQL, Oracle (PL/SQL), MongoDB
- **Web & Backend:** Node.js, Express.js, Next.js, API development, Git
- **Security:** Web pentesting (OWASP Top 10), threat hunting, infrastructure bug testing
- **Embedded:** Embedded software development, hardware integration, RTOS, embedded security
- **Languages:** Python, Bash, C, C#, C++, Java, YAML

PROJECTS

- **AI Language Tutor (in progress):** Prototype of a private AI tutor for language learning, built on an open-source base model with a RAG pipeline and vector database. An LLM Factory lets it run on OpenAI APIs or local models through Ollama (Llama 3, Mistral); a Vector DB Factory abstracts storage, with groundwork laid for Postgres pgvector. Documents are ingested through a FastAPI service, embedded and indexed for semantic search. Product vision: conversational practice, persistent progress records, and future certified exams.
- **CI/CD Pipeline:** Built an end-to-end GitOps pipeline (GitHub Actions → Docker → Argo CD → Kubernetes) for a containerized microservice application, automating both integration and deployment stages.
- **Web Portfolio:** Automated a serverless deployment pipeline for a high-performance Next.js portfolio, using Terraform for cloud infrastructure provisioning and GitHub Actions for continuous delivery. GitHub
- **ADAS, Graduation Project:** Advanced Driving Assistance System with five subsystems for collision avoidance. Distance, image and timing inputs from external hardware are processed by embedded software on a Raspberry Pi, and a V2V connection lets both vehicles make decisions when an event is detected.
- **Bookstore Backend (personal project):** Backend application built with Express.js and MongoDB on Node.js. GitHub

CERTIFICATIONS

- Red Hat System Administration II (RH134)

LANGUAGES

Arabic (Native) English (Proficient)